

Perspectives of resilience in mega-event studies in the context of urban development: A systematic literature review

Longmiao Gao^a, Yan Liu^{b,*}, Fiona X. Yang^a, Huimin Liu^b, Ying Wang^c

^a Faculty of Business Administration, University of Macau, Avenida da Universidade, Taipa, Macao

^b School of Management and Engineering, Nanjing University, 5 Pingcang Lane, Gulou District, Nanjing 210093, PR China

^c School of Hotel and Tourism Management, Hong Kong Polytechnic University, 17 Science Museum Road, TST East, Kowloon, Hong Kong

ARTICLE INFO

Keywords:

Resilience
Mega-event
Systematic literature review
Urban development

ABSTRACT

Mega-events have been considered as a particular form of mega-development projects that can influence the resilience of urban systems and the environment. Although building resilience has become a priority for event management following the creation of event legacy and sustainability, the current application of resilience in mega-event research remains unclear in terms of definitions, issues, and an overarching overview. This paper adopts a systematic literature review, first examining various types, descriptions, and definitions of resilience related to mega-events, such as urban resilience and regional resilience. Difficulties in applying resilience concepts in current mega-event research are highlighted, considering definitional, descriptive, situational, and objective ambiguities. Furthermore, relevant research in the past two decades has gradually revealed the uncertainties and vulnerabilities in mega-events and their development processes in different scenarios. Therefore, this paper summarizes five key themes related to resilience and mega-events: a) urban regeneration and environmental management, b) infrastructure and transport development, c) regional economy and tourism development, d) local community adaptation, and e) safety and security commitment. Given the challenges and risks arising from the intersection of event management and urban development, this paper advocates an overview framework to guide the application of resilience perspectives in mega-event research, providing a theoretical and literature foundation for further exploration of resilience issues in future mega-event research.

1. Introduction

In recent decades, city authorities have sought constructions and developments led by mega-events as opportunities to introduce urban regeneration plans (Girgin & Tasci, 2019). Prominent mega-events such as the Olympic Games, World Expos, and World Cups are considered to bring “once-in-a-lifetime opportunities” for cities and communities in a short time (Thompson et al., 2013). It is often assumed that the success of a mega-event can regenerate the so-called “problem areas” and create sustainability, resilience and inclusion for the host city (Raco & Tunney, 2010). However, these event-led large-scale development projects have also been proven to create enormous challenges for local governments, residents, and the environment (Gezici & Er, 2014; Robinson & Torvik, 2005). Emerging literature highlights a series of issues encountered in event-led urban development with increasing urgency, such as climate change (Scott et al., 2019), terrorist attacks (Coaffee, 2021), financial

crisis (Robinson & Torvik, 2005), social injustice, spatial fragmentation (Azzali, 2019), inequality (Rocha & Xiao, 2022), global epidemics (Lee Ludvigsen, 2021), etc. These risks have led to a shift in the discussion of mega-events from short-term impacts on economic growth, tourism increment, and city image to long-term, resilient transitions from the host of mega-events to cities' event-led urban development programs and agendas (Darnell, 2012), especially considering mega-events interactions with local social, economic, and environmental ecosystems (Shin & Welty Peachey, 2022).

Therefore, mega-event literature has highlighted the term ‘resilience’ in the past two decades (Chen et al., 2021; Girgin & Tasci, 2019; Orr et al., 2022). Resilience was originally used to address transportation challenges and terrorist threats in socio-spatial territories, mainly outbreaking during mega-events (Bovy, 2006; Toohey & Taylor, 2008). Gradually, resilience spreads to other urban development aspects, such as sports venue construction, infrastructure renewal, tourism

* Corresponding author.

E-mail addresses: yc47014@um.edu.mo (L. Gao), ly@nju.edu.cn (Y. Liu), fionayang@um.edu.mo (F.X. Yang), liuhm@nju.edu.cn (H. Liu), yings.wang@polyu.edu.hk (Y. Wang).

<https://doi.org/10.1016/j.cities.2025.106058>

Received 6 July 2024; Received in revised form 28 October 2024; Accepted 29 April 2025

Available online 9 May 2025

0264-2751/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

stimulation, urban spatial transformation, economic restructuring, etc. (Richards & Wilson, 2004; Shin, 2014). Recently, the Tokyo Olympics' rebirth and recovery in a post-pandemic world symbolized the significance of resilience, which was highly recognized by the IOC (International Olympic Committee) (Yamamura, 2021). In the urban context, resilience emphasizes enhancing the capacities of the urban (sub)systems and their subjects to transform, adapt, and continuously bounce forward and evolve towards new states (Meerow et al., 2016). Therefore, it is necessary to understand how event and city planners and managers prepare, adjust and resolve to create a resilient and future-proof urban system and how cities and citizens anticipate, respond and co-evolve in the event-led development process (Deakin et al., 2019).

Mega-events tend to be regarded as singular points in time, often downplaying the long-term urban development process that precedes any presumed benefits to local people from the event (Clark et al., 2016). Scholars have pointed out that the contradiction between planning discourse in mega-events and the high degree of uncertainty and complexity in the environmental conditions brings vulnerabilities (Li, 2021; Rocha & Xiao, 2022). In this context, resilience perspectives align with the new trend of 'glocalization' in mega-events (Shin & Welty Peachey, 2022), i.e., bringing greater foci on vulnerabilities and adaptability in event-led urban development, along with capacity building for urban systems and social agents. Therefore, the goal of this paper is to construct a dialectical and nuanced view of resilience as an emerging discourse of mega-events impact and to present it in a practically discussable and neutral manner based on reviewing and integrating the dispersed and segmented current research. Rather than exploring single risk factors, this paper focuses on integrating the challenges and opportunities associated with resilience in mega-events both in the long and short term, endogenously and exogenously. This review thus aims to answer the following research questions:

How has the concept of resilience been contextualized and applied in mega-event studies?

What aspects of mega-events (including event-led urban development processes) have the resilience concept been applied to?

How can we establish a framework to guide the application of resilience in future mega-event research?

2. Methodology

This systematic literature review (SLR) is mainly based on the PRISMA method (Moher et al., 2015). The initial search retrieves publications from two databases, Scopus and Web of Science, which are commonly used sources. Three sets of keywords are defined. The first set of keywords is: "resilien*" AND ("mega-event*" OR "mega event*" OR "mega-event*" OR "mega sport* event*" OR "hallmark event*" OR "major event*" OR "large-scale event*" OR "cultural event*"). The formation of the second set of keywords ensures the research of three commonly studied mega-events, i.e., "resilien*" AND ("Olympic*" OR "Expo" OR "World Cup*"). The third group of keywords, "event-led" AND "urban development," is also included. Keywords were used in the subject, title and abstract fields.

2.1. Data searching

We initially included all online publications as of January 2024, including initial database search hits 397 on Web of Science and 381 on Scopus. Then, duplicate entries were removed. The e-books, book chapters, conference articles, and reports found in the initial search were excluded to focus on peer-reviewed academic journal articles published in English (Cronin et al., 2008). After screening the titles and abstracts, unrelated articles were excluded. Moreover, articles on irrelevant events (natural events, specific computer terms, etc.) were excluded in research fields such as computer science, psychology, behavior sciences, and educational research. After this round of screening, the remaining 68

articles entered the next step.

2.2. Sample selection

In the following stage, the first author read the remaining studies thoroughly to analyze each article. Articles that lacked any of the following inclusion criteria were excluded: articles that emphasized resilience in mega-events or specific events; articles that addressed resilience or specific capacities related to mega-events; articles that provided a view of the urban context of mega-events; and articles that provided a background in resilience theory. Afterward, the authors conducted snowball sampling and added 13 articles to ensure a more comprehensive data collection. Following the above standards, the final sample includes 49 articles within the research scope.

2.3. Data analysis

The third stage of the review includes analyzing and reporting the results. All the articles were analyzed with the following information: importance/relevance to the topic, classification, topic/theme/issue, academic area, theoretical lens, publication year, journal, methodology, type/definition/context of resilience, research question, main finding, case, central tension, etc. The results are discussed below, with the findings drawing on several 'W-questions' for resilience in general, i.e., the resilience of what, resilience for what (for when, where and whom), and resilience to what (Meerow & Newell, 2019). Figs. 1 and 2 present the distributions of the reviewed articles by year of publication and journal source.

3. Findings

3.1. Resilience of what? Descriptions and definitions in mega-event studies

Mega-event researchers have applied the terms resilience/resilient/resiliency in different scenarios, characterizing resilience connotations from different perspectives or adopting definitions available in other research fields. However, not all included sample articles elaborate on the definition or implications of resilience. Only a portion of the samples provided descriptions, and a few adopted detailed definitions, reflecting the reliance on a consensus around an implicit yet unclear meaning of resilience in current literature. To address the issue of overly vague definitions in the current research, this section summarizes 22 of the 49 samples that contain descriptions related to the definition of resilience and clusters them based on existing academic definitions used by researchers, deriving five definition categories: urban resilience, infrastructure and transport resilience, regional and tourism resilience, community resilience, and security resilience.

3.1.1. Urban resilience

Urban resilience refers to the capacity of urban systems and their subsystems to recover and adapt to crises in the face of constantly changing environments, as well as unexpected and unpredictable disturbances and shocks. This concept views cities as complex adaptive systems involving interrelationships among multiple actors (Meerow et al., 2016; Yamamura, 2021). Mega-events are special urban development projects that have various impacts on urban systems, and their developments constantly alter tangible and intangible environments and foster interactions among stakeholders and components. Three sample articles mention three types of "disturbances" to maintaining urban resilience: perturbation and interactions between the involved agents (Salvati & Serra, 2016), shocks/crises to security and public safety (Boyle, 2012), and air pollutants (Feng et al., 2022), which shows that the current mega-event literature uses this concept across various topics. According to Yamamura (2021), to create urban resilience in event management, event organizers and managers should strive to address various disturbances and ensure permanent positive environmental

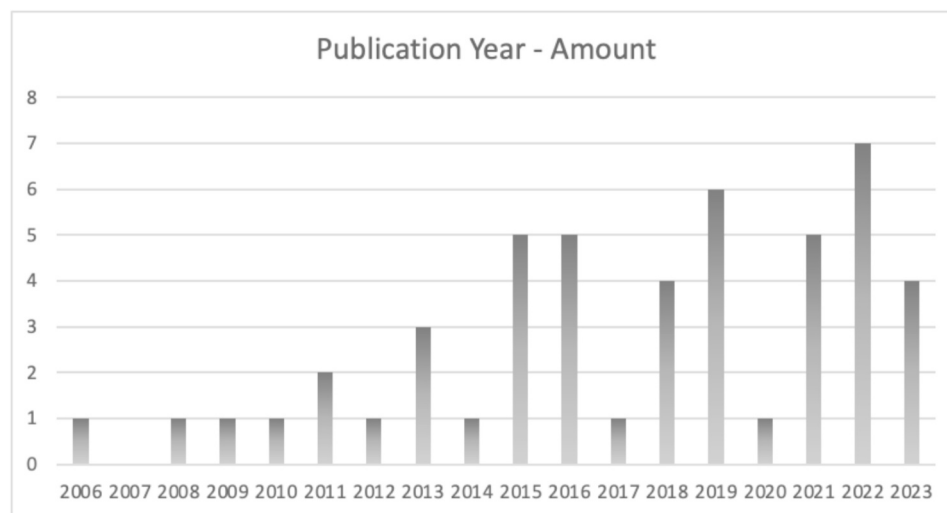


Fig. 1. Articles by year of publication.

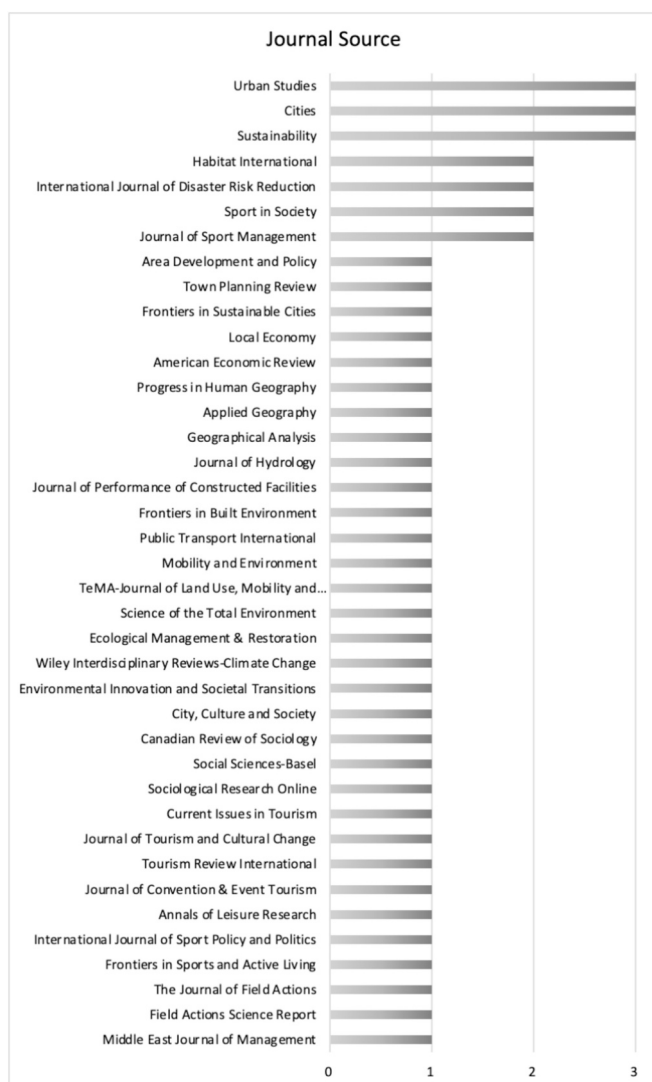


Fig. 2. Journal sources of selected literature in the findings.

changes. Urban resilience encompasses different types of urban systems with their prioritized issues, causing thematic overlaps of its application subjects with other types of concepts (e.g., regional and security resilience). Therefore, relying solely on the metaphor of urban resilience without in-depth discussion and overlooking such diverse topic scenarios can hinder future research. According to different subjects and objectives, Boyle (2012) emphasizes urban systems' adaptation and rebound, while Feng et al. (2022) describe it as restoring and recovering rapidly, i.e., prompting the urban system to either move forward or return to its original state (Shipway & Miles, 2020). Building urban resilience in event-led urban development requires breaking down ineffective planning and governance failures and changing unbalanced development patterns and path dependencies (Goodwin et al., 2019).

3.1.2. Infrastructure and transport resilience

Infrastructure such as transport can be seen as subsystems of the urban system, and mega-events can impact their resilience, e.g., by focusing on the resilience of urban logistics networks and their recovery plans in supply disruptions or congestion caused by mega-events (Zheng et al., 2023). Bovy (2006) stated that transportation for mega-events has historically posed a severe and significant operational risk. Some Olympic Games brought increased transportation supply and infrastructure investment in developing countries and megacities, where demand for transportation was huge, trip-lengthy and in poor socio-economic conditions (Ferranti et al., 2022). Ross and Orr (2022) describe resilience as the integration of climate adaptation and resilience measures within infrastructure development for mega-events. The contradiction between mega-event priorities and the mid-and long-term requirements of urban transportation systems presents an operation challenge (Ferranti et al., 2020). Serdar and Al-Ghamdi (2021) stated that ill-planned transportation expansions not only cause degradation in service quality but also add more loads to the transport systems, considering the rising frequency of disasters affected by political tensions, climate change, etc. Host cities must plan and prepare for the long-term development of transportation systems to protect an exponentially growing number of visitors during and after the event from potential natural disasters such as earthquakes (Nguyen et al., 2019).

3.1.3. Regional and tourism resilience

Regional resilience is concerned with regional socio-economic structures affected by the environment, with mega-events serving as enablers for urban growth. In current mega-event research, regional resilience refers to the ability of certain cities and metropolitan areas to reconfigure their socio-economic structure of industries, technologies,

institutions and firms to adapt to change (counter maladjustment) and cope with demands, thus maintaining an acceptable growth path and constituting the material conditions for developments (Deakin et al., 2019; Lin et al., 2018; Ramos & Mundet, 2021). Challenges to regional resilience posed by mega-events include the overexploitation of the city due to looming local debts and liabilities, chronic over-tourism development, and commodified and overcrowded public spaces, leading to gentrification, regional inequalities, and social stratification, e.g., the expulsion and marginalization of residents from their neighborhoods (Ramos & Mundet, 2021).

Compared to regional resilience, tourism (or destination) resilience focuses more on the impact of tourism development on the socio-economic ecosystem of destinations under mega-event strategies. Tourism resilience in the context of mega-events implies that destinations should better cope with new changes in governance structures and their impacts as a result of the event, as well as address current and future economic, social, and ecological crises and threats (e.g., terrorism, natural disasters, pandemics, climate change). In this way, destinations can maintain new growth pathways and build long-term capacity so that the event enables tourism to deliver long-term benefits (Chen et al., 2021; Lee & Hlee, 2021). In the long run, coordinating and mobilizing assets among various stakeholders in the event's destination region are vital to building tourism resilience.

3.1.4. Community resilience

While the literature often emphasizes the impact of mega-events on local communities, relatively few studies focus explicitly on the resilience of local communities. The post-use and capital repayment of mega-event venues and infrastructure are questioned, as they may impose long-lasting stresses on local societies and communities in restoring local production after the event (Lee, 2019). Therefore, Nakamura and Suzuki (2017) understand the resilience of local communities as the capacity of host communities to cope with the inevitable long-term nature of mega-event legacies and their unintended consequences. Even years after an event, host communities need to reconfigure event legacy assets to achieve long-term planning and development goals (Michelam et al., 2021). This process requires clear strategies for the post-event period, designed and planned with the support and commitment of local and political stakeholders, to build community resilience through continuous learning and reinterpretation (Bernardino et al., 2018; Nakamura & Suzuki, 2017). Rocha (2023) contributes a different perspective on community resilience, describing it as fostering ties of communities to build resilience in the face of tragedies and disasters caused by events such as evictions. This bottom-up perspective enriches the interpretation of community resilience by providing a relative discussion on collaboration and resistance in the negotiation process of a mega-event with the community rather than just being confined to the grand narrative of event-related constructions and development (Deng et al., 2016).

3.1.5. Security resilience

Security (or safety, venue) resilience focuses more on safeguarding the functioning of infrastructure systems in hosting mega-events and on capabilities to respond to terrorism. Mega-events bring together large numbers of populations in the host city in a short period, concentrating risks in particular areas and leading to increased severity of environmental, social, and political consequences. Girgin and Tasci (2019) define venue resilience as the capacity to withstand disruptions to infrastructure and services to safeguard the functioning of a mega-event, as well as to ensure the security and safety of infrastructure, addressing a broad range of risks caused by human and environmental factors. Some articles are devoted to addressing the risk of terrorism and terrorist attacks during mega-events, particularly in the aftermath of the 9/11 attack (Rekow, 2015). In this context, security resilience means developing a solid strategic capacity by security managers and policymakers to proactively plan counter-terrorism operations and enhance resilience

to potential terrorist attacks through an urban emergency planning approach (Coaffee et al., 2011). However, such “exceptional” actions may exacerbate spatial segregation, leading to urban fragmentation and geographic disconnection.

3.2. Resilience for what (when/where)? Thematic analysis of 5 thematic areas

Section 3.1 focuses on summarizing several concepts of resilience used in mega-event research based on a portion of sample articles. Although other samples do not provide clear definitions, their research still focuses on issues related to resilience. Therefore, this section analyses the five thematic areas generated from sample articles, considering their specific resilience issues. It further situates the discussion of resilience within various mega-event and urban development scenarios regarding time, location, actions, stakeholders, etc.

3.2.1. Urban regeneration and environment management

3.2.1.1. Urban regeneration. Mega-events function as catalysts for adaptive evolution and resilient transition of urban systems, bringing substantive urban fabric changes that might otherwise take decades, leading to the spatial transformation of host cities and surrounding regions (Deng et al., 2016; Thompson et al., 2013). The 2012 London Olympics changed “London's worst poverty” by cleaning up over 300 ha of century-old industrial pollution, synergistically integrating Olympic space demands and the transformation plan of the core city (Thompson et al., 2013). Similar urban renewal trajectories are evident in other cases: the 2004 Athens Olympics saw downtown land use patterns become culturally creative and technology-intensive (Gospodini, 2009); the 2022 Beijing Winter Olympics completed the post-industrial heritage transformation of Shougang Park (Deng et al., 2020); and Expo Milano 2015 converted the urban brownfield into an innovation district called “MIND,” attracting startups and fostering vibrant communities (Deakin et al., 2019). While mega-events have emerged as effective tools for fostering resilience in the post-industrial revival of problematic areas, they present a singularity paradox between temporary projects and permanent spatial transformation (Grabher & Thiel, 2015). For example, the demolition of old traditional building blocks (hutong districts and courtyard houses) during the 2008 Beijing Olympics changed the city's historical fabric and caused the residents of these neighborhoods to leave familiar living spaces (Gu & Cook, 2012; Wang et al., 2015). Decisions lacking a long-term vision may lead to conflicting urban development objectives across different spatiotemporal scales, resulting in uneven distribution of resilience (Gaeta & Vita, 2021).

3.2.1.2. Climate adaptation. Climate change imposes increasingly stringent constraints on mega-event host cities. Scott et al. (2015) find that while 19 cities hosting the Winter Olympics between 1981 and 2010 had suitable climates, only six cities may remain viable options by 2080. Using the Australian Open as a case, Gollagher and Fastenrath (2023) introduce a transformative climate resilience framework that explores the potential of large-scale sports events to collaborate with stakeholders in advancing urban climate adaptation. Orr et al. (2022) examine climate change impacts on the operation of competitive sports entities and their adaptive measures. These studies highlight the necessity for cities to develop proactive climate-resilient adaptive capacity in preparation for future environmental challenges to long-term legacy planning, such as winter sports facilities and venues.

3.2.2. Infrastructure and transport development

3.2.2.1. Transport development. The value of sustainable and resilient transport legacies of mega-events has become prominent (Pereira, 2018). Infrastructure and transport considerations for mega-events have

shifted towards prioritizing a resilient and robust infrastructure network as a foundational requirement to meet transport demands (Bovy, 2006) alongside the comprehensive improvement of resilient networks of interlocked infrastructures (Serdar & Al-Ghamdi, 2021), considering the dramatic increase of natural disasters (e.g., climate change and flood) and human factors (e.g., political tensions, accidents, and rapid urbanization). For instance, Serdar and Al-Ghamdi (2023) evaluate post-flood interventions for the 2018 Qatar World Cup. Given the high-intensity disasters suffered, Japanese railway companies developed an infrastructure risk contingency plan for the 2020 Tokyo Olympics, enhancing the preparation of transport systems and society to recover from possible hazards (Nguyen et al., 2019).

Nonetheless, there are still tensions between the short-term goals of event-oriented construction and the long-term needs of urban transport systems. Due to this point, the bus rapid transit system of the 2010 FIFA World Cup in South Africa and the 2016 Rio Olympics face long-term financial and operational challenges amid future extreme weather after the event (Kobierecki & Pierzgański, 2022). Future research can examine cascading effects among various infrastructure types, and ill-planned infrastructure expansion needs to be especially vigilant (Chopra et al., 2016).

3.2.2.2. Resource systems development. Mega-events can serve as catalysts for promoting the restoration and adaptability of urban resource systems and supply chains. In recent years, the World Expo has advocated a highly forward-looking environment and resource strategies for facing global challenges. For instance, smart food and climate response systems are significant outcomes of the Expo (De Angelis et al., 2016). The slogan of Expo Milano 2015 is “Feeding the planet, Energy for life” (Gaeta & Vita, 2021). Eco-friendly economic and innovative behavioral modes support institutions in resource utilization and regional growth (Deakin et al., 2019). In the 2012 London Olympics, urban resilience was increased by addressing the governance challenges of the non-potable water reuse scheme through stakeholder engagement (Ross & Orr, 2022). Mega-events thus bring opportunities for building resilient governance structures within urban resource systems.

3.2.3. Regional economy and tourism development

3.2.3.1. Regional economy. Mega-events are critical developmental hallmarks in regional political economy (Gezici & Er, 2014). In applied economic modeling, they can be considered fluctuating factors with economic implications influencing regional economy resilience. Salvati and Serra (2016) demonstrate how the 2004 Athens Olympics brought rapid system-level changes that facilitated the metropolitan area, as a complex adaptive system, to enter a new evolutionary stage, supporting resilience-promoting policy designs in the region. Through multi-case analysis, Lin et al. (2018) emphasize that mega-events are momentary interventions to prolong regional economic development cycles. However, they may undermine regional resilience by aggravating social inequality and stratification under the repositioning of stakeholders.

3.2.3.2. Tourism development. Mega-events increase visitor arrivals and provide opportunities for business leverage, destination branding, and induced tourism growth (Wang & Jin, 2019). Mega-events can function as place-branding strategies for host cities, fostering more resilient stakeholder networks and promoting entrepreneurial urban development (Edizel, 2013). However, misalignment between entrepreneurial practices and growth politics can have disruptive forces and create conflicting governance agendas (Lauermann, 2016). Decades after the 1988 Barcelona Olympics, the over-deprivation from the tourism industry, weakening socio-economic structures, real estate speculation and resocialization, and gentrification of public spaces have become everyday situations in the city, described as “tourism-phobia” (Ramos & Mundet, 2021). Due to the lack of long-term resilience after tourism

growth, developmental imbalance within rapidly remodeled industrial structures is often overlooked, frequently disregarding the interests of diverse sectors and stakeholders.

3.2.4. Local community adaptation

Although community resilience is less defined in the literature than other concepts, practical issues frequently appear in existing studies, such as spatial inequity and gentrification (Rocha & Xiao, 2022). The rapid rise of certain industries, businesses, and communities through mega-events in particular regions often comes at the expense of space, resources, and development opportunities for disadvantaged and marginalized stakeholders in peripheral areas (Grabher & Thiel, 2015). The lack of consideration of residents' interests in land upgrades and built environment expansions in mega-events usually harms the resilience of pristine communities (Lenskyj, 2012). For example, the 2014 Glasgow Commonwealth Games initially claimed to create social housing; however, due to rising land prices, a large part of the converted housing ended up supplying to temporary investors and migrants seeking new opportunities (Clark et al., 2016). Some have argued that mega-events can be abusive tools used to accelerate land sales, increase debt leverage, and transfer populations, revealing the controversial relationship between state power and spatial reproduction (Lin et al., 2018). Although the 2016 Rio Olympics aimed to implement community-based micro-improvement plans before its hosting to alleviate the pain and insecurity of slum residents (Rekow, 2015), Rocha (2023) found that the event led to expulsions and tragedy, and people had to strengthen ties between communities to resist and stay in their communities.

Opportunities are also open to creating resilient spaces within overstressed communities (Lauermann, 2016; Rekow, 2015). Beyond public and private sectors, how individuals, communities and social groups access resources, adapt over time and build capacities is critical, such as through post-demolition resettlement and job training (Boyle, 2012; Chen et al., 2021; Rogerson, 2016). Recent literature describes the practices of host sites in helping microenterprise communities adapt to local economic restructuring and enhance local resilience. Nakamura and Suzuki (2017) recall that since the 1998 Nagano Winter Olympics, despite initial regional damage, Hakuba's government has gradually optimized its governance system through strategic planning, helping build soft and hard facilities conducive to the adapting, learning, and transferring experience of local enterprises over the past two decades.

3.2.5. Safety and security commitment

Mega-events require a solid commitment to safety and security to demonstrate resilience. In the aftermath of 9/11, the 2004 Athens Olympics explored comprehensive approaches to prepare for and manage terrorism risks (Toohey & Taylor, 2008). Security preparations for the 2010 Vancouver Winter Olympics improved the “whole-of-government join-up governance” of the city, providing unique opportunities to plan for unexpected and uncertain vulnerabilities (Boyle, 2012). The success of the 2012 London Olympics in dealing with security risks stemmed from the wake-up call when London's public transport network was attacked in July 2005, shortly after winning the bid (Draca et al., 2011). Coaffee (2021) regards resilience-building as a transformational process, and the security of the London Olympics is a drill in flexibility and adaptability. Literature on mega-event security also emphasizes fostering learning and knowledge-sharing within the stakeholder network (Girgin & Tasci, 2019). However, scholars also question whether the emphasis on single-dimensional risks to security is an omission of systemic fragility, specifically whether the presuppositions of security priorities create unintended inequalities, such as the monitoring and geo-restrictions imposed on residents through security measures for mega-events (Boyle, 2012; Clark et al., 2016).

4. Discussion

4.1. Challenges in applying resilience concept in mega-event studies

This section analyses some challenges in applying the concept of resilience in mega-event research regarding the definitional, descriptive, situational, and objective ambiguities discovered in the findings and proposes some suggestions.

4.1.1. Definitional ambiguity

A considerable portion of the current research lacks a precise definition and instead uses resilience as an ambiguous modifier. According to Fröhlich and Hassink (2018), due to the polysemous nature of the resilience concept, academics have questioned whether resilience is a “stretched” concept. Given the wide variation in specific research objects (urban subsystems) of resilience, relying solely on academic consensus to interpret its meaning - without a precise definition - can lead to ambiguous and misleading understandings. We suggest that further research on developing resilience concepts in mega-events should normalize and clarify their definitions.

4.1.2. Descriptive ambiguity

Greater clarity is needed when describing the resilience concept in mega-event research. Much of the literature borrows existing definitions of resilience concepts (e.g., urban resilience) without adaptation. More detailed descriptions are required to apply this concept accurately. Some studies equate resilience with concepts like “adaptive capacity” and “robustness,” while others use loose descriptions such as “resilience for coping with...[some listed crises]”. Meerow and Newell (2019) argue that clarifying resilience requires addressing a series of questions of resilience for whom, what, when, where and why. Specifically, it is insufficient only to emphasize adaptation; it is necessary to specify who is transforming and adapting, what challenges are being addressed, etc. The absence of these essential descriptors complicates the development

of a coherent definition and often requires inferences beyond the text.

4.1.3. Situational ambiguity

There is often a lack of clarity regarding the scenarios in which the resilience concept is applied, as evidenced by the less correlation between the concept description and the specific disturbances it references or the challenges it addresses in a mega-event context. Whether the challenges faced by the (sub)systems are long-term or short-term, present- or future-oriented (in mega-events or the event-led development), endogenous (brought about by mega-events) or exogenous, may result in a different interpretation of resilience. Future research should be wary of the mismatch between the conceptual perspectives of resilience, its application scenarios, and the challenges to be addressed in specific mega-event scenarios, which may bring about ambiguity between academic concepts and practical issues.

4.1.4. Objective ambiguity

Finally, a critical consideration is a trade-off between the specificity and generality of ‘resilience for what.’ Resilience can be used to respond to and recover from specific threats associated with mega-events (e.g., climate adaptation or terrorist attacks), or it can emphasize transforming the overall system in preparation to respond to and adapt to a range of uncertain and unexpected disturbances in event-led urban development. It is necessary to distinguish, trade-off, or integrate goals in building mega-event resilience, as different directions may lead to varying academic priorities and practical contributions.

4.2. An overview framework of resilience in mega-event studies and its application

Based on the previous analysis, this section synthesizes an overall framework of resilience in mega-event research (Fig. 3, including four modules), intended to help further understand this research field and guide the appropriate application of resilience in future studies,

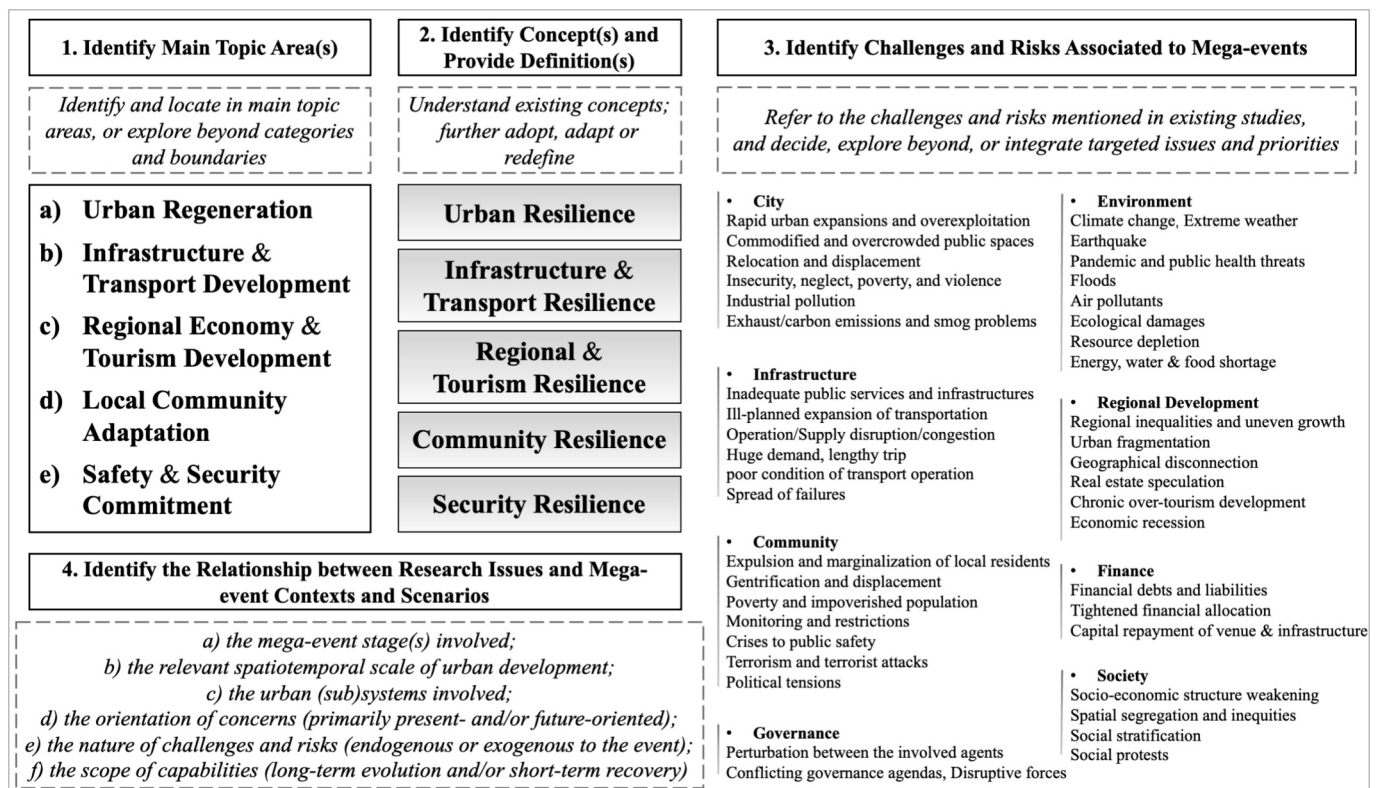


Fig. 3. An overview framework of resilience in mega-event research.

avoiding the ambiguities mentioned earlier. The first and second modules show thematic areas and definitions summarized through the above review. It is worth noting that they are not necessarily in a one-to-one relationship within the sample articles, although there are instances of overlap. For example, articles studying infrastructure and transportation issues may use the concept of urban resilience. These two modules represent the niche and conceptual choices that need to be clarified based on specific research questions, considering their relationship as well. Some critical inquiries may follow: If focusing on one topic (e.g., infrastructure), which resilience concept should be chosen (urban resilience, infrastructure resilience, or another beyond this category)? Why choose this type, and how should it be defined? In addition, when research spans multiple domains and crosses category boundaries, how do researchers select, integrate or combine relevant definition(s)? Should an analytical framework be established by integrating several relevant dimensions?

While mega-events often focus on ambitious goals set in advance (Horne & Manzenreiter, 2006), overlooked challenges and risks can hinder their fulfillment, resulting in mismatches between actual implementation processes and promised attractive outcomes in mega-event practices. Therefore, in the third module of Fig. 3, the framework summarizes the resilience-related challenges and risks identified from the literature, which still do not necessarily have fixed relationships with the thematic areas and/or definitions due to a lack of clearly articulated links in the literature. We further clustered them into groups for future research reference, including city, environment, infrastructure, regional development, finance, community, society, and governance. Importantly, these categories serve as reference points rather than rigid classifications, allowing researchers to select, combine, or expand beyond them based on specific research needs.

Moreover, we suggest that the cascading and spillover effects of risks across different urban subsystems require attention when concerning resilience. For instance, extreme weather may impact infrastructure and visitor security, while regional economic restructuring can influence community and inhabitant well-being. Measuring and monitoring only a specific set of indicators may limit a complex system's approach to resilience (Quinlan et al., 2016), as current literature emphasizes recognizing the impactful corruption of "slow-burning" crises on systems beyond a single shock (Hassink, 2010). Resilience priorities and arrangements in one domain may generate unexpected impacts on priorities in other domains, and such spillovers may increase vulnerability at a systematic level. For example, security restrictions may lead to dissatisfaction in local communities during mega-events (Rocha, 2023). These interconnected priorities and actions are often difficult to coordinate and manage effectively, posing substantial governance and operational challenges. Tensions between resilience goals can be compounded by the mismatch of event agency requirements and immediate local demands (Müller, 2015), the political urgency of hosting mega-events (Kassens-Noor & Lauermann, 2017), and the imbalance between short-term event needs and long-term legacy and city plans (Talbot, 2023).

The fourth module of Fig. 3 addresses the question of "resilience at what scale?" (Weichselgartner & Kelman, 2015). When resilience, a broad concept, is applied to mega-events and host cities as complex, evolving systems, it is unrealistic to rely on inference alone to determine the implicit relationship between the proposed resilience-related issues and specific mega-event contexts and scenarios. To avoid contextual and objective ambiguities, future research should explicitly delineate the precise relationship, and may follow and address several key points listed in the module: a) the mega-event stage(s) involved; b) the relevant spatiotemporal scale of urban development; c) the urban (sub)systems involved; d) the orientation of concerns (primarily present- and/or future-oriented); e) the nature of challenges and risks (endogenous or exogenous to the event); f) the scope of capabilities (involving long-term evolution and/or short-term recovery), etc. For example, while mega-events are easily viewed as discrete points in time (Clark et al., 2016),

their development trajectory encompasses multiple phases (bidding, preparation, staging, and post-event developing) and geographical scales (venues, core areas, the peripheral areas, and widely radiating regions).

4.3. Linking resilience to ongoing debates in mega-event studies

4.3.1. Resilience, sustainability, and heritage in mega-event studies

The topic of resilience has not been at the center of major discussions of mega-events, and previous research on the impacts of mega-events has mainly covered sustainability and heritage, including a few reviews (Meza Talavera et al., 2019; Thomson et al., 2019). Recently, Müller et al. (2021) conduct systematic evaluations on the sustainability of the Olympic Games as iconic mega-events. They point out that despite these events' strong promotion of hosting sustainable Olympic Games, the reality was not the case. Scholars criticize that mega-event planning discourses have been seen to overstate positive future visions, soften negative features, and ignore potential risks (Duignan et al., 2022). In this case, the resilience concept introduced in our paper is complementary to sustainability, which usually acts as a goal. Resilience thinking inspires attention to the uncertainty, nonlinearity and vulnerability in the planning and governance of urban mega-event projects and advocates a continuous adaptative transformative paradigm. In the editorial paper on mega-events and heritage, Ponzini (2022) proposes that how to combine mega-events and heritage to demonstrate greater flexibility and cope with future restrictions is a challenge. It's indicated that the main issue for the medium- and long-term development of mega-events and cities involves urban heritage. According to Nakamura and Suzuki (2017), long-term evaluation of post-event strategic planning should be encouraged to understand the resilience of host communities in dealing with the changing role and inevitably long-term nature of event legacies.

4.3.2. Resilience and governance in mega-event studies

The debate on mega-event resilience is also relevant to governance issues in event-led urban development. While research highlights the importance of taking proactive actions in urban governance dynamics and changes triggered by mega-events (Majoor, 2015; Silver, 2014), enabling resilience through governance transformation and institutional capacity building faces significant practical challenges. For instance, Boyle (2012) advocates "whole-of-government joint governance" to achieve security resilience. However, its implementation in practical situations still faces many obstacles, including established bureaucratic procedures, competing sectional priorities, and complex power dynamics among stakeholders.

Nevertheless, efforts are still needed to develop the "resilient" features in the governance mechanisms and processes of mega-events, which is related to conceptualizing resilience in terms of organizational networks, governance systems and long-standing partnerships in mega-events (Gundes, 2022). This highlights the importance of considering resilience as a systemic property when designing and testing resilience frameworks in multi-actor networks and partnerships of government agencies, civil society and private sectors (Weichselgartner & Kelman, 2015). The obstacles to resilience-oriented governance suggest that it is still needed to develop the resilience of governance mechanisms, systems and processes themselves, to provide governance support for the development of resilience capabilities.

5. Conclusions

While mega-event organizers justify substantial expenditures by emphasizing long-term structural changes for host regions, critics have noted that mega-events can shift many of the problems in urban growth elsewhere and postpone crises into the future (Flyvbjerg et al., 2016; Lin et al., 2018). In this context, resilience can be a valuable framework for strengthening and improving systems (Flax et al., 2020), which applies

to mega-events and urban systems associated with them, such as the venues and surrounds, and the broader host city (Coaffee et al., 2011). The resilience paradigm particularly provides event managers, city planners and stakeholders with an approach to anticipate, respond to and adapt to dynamic changes, unforeseen circumstances and unexpected challenges (Chen et al., 2021; Kwiatkowski et al., 2023). However, as Walker (2020) cautions, resilience risks becoming a fleeting buzzword: “Sometimes it is open to interpretation, and sometimes it is simply wrong.” While resilience has attracted growing attention from mega-event researchers and practitioners across various domains, the absence of a systematic summary and integration leaves its application in mega-event contexts ambiguous. Our paper, therefore, investigates the current application status of resilience concepts in mega-event research through a systematic literature review, analyzing different resilience-related concepts and the topics discussed in current mega-event research across disciplinary boundaries. This paper also contributes to the understanding and development of resilience and its application by uncovering the concept in the field of mega-events.

This paper finds that scholars have adopted various types of resilience terminology. As our analyses demonstrate, these resilience terms are ambiguous in their definitions, descriptions, and application scenarios, causing these concepts to be prone to confusion, with difficulties in clarifying their connotations and identifying their boundaries and overlaps with others. We find that the relatively vague or absent definitions of resilience in mega-event research, with intertwining underpinned meanings, make it challenging to identify and analyze practical mega-event issues, hindering clearer and deeper theoretical analysis.

Another contribution of this paper is reviewing the thematic areas in mega-event studies related to resilience, categorized as urban renewal, infrastructure and transport development, environmental management, regional economy and tourism development, local community adaptation, and safety and security commitment. However, this part of the analysis is, to some extent, limited by the predominance of single-disciplinary knowledge, as the majority of the current literature exists within separate disciplinary frameworks. We suggest that future mega-event research expand the scope of resilience and appropriately adjust the boundaries of urban systems affected by these events, particularly given that investigating the causes and solutions to existing issues requires cross-disciplinary knowledge from and cooperation with scholars and practitioners.

This paper then proposes an overview framework for resilience in mega-events and event-led urban development. We emphasize that future mega-event studies on resilience should focus not only on adapting to specific threats but also on building resilience in the long-term evolutionary processes of the overall urban system, addressing risks and challenges with a futureproofing approach. Particularly in the post-pandemic era, uncertainty and vulnerability pose greater challenges to the collision of mega-event management and urban governance. The structural classification and overall framework provided in this paper can serve as a theoretical foundation for both academics and the industry to further investigate the management and governance of mega-event resilience in the future.

A limitation of this systematic literature review is its focus exclusively on “peer-reviewed academic journal articles”, which may constrain the review’s comprehensiveness by omitting influential books and book chapters. Future research could expand this dataset to include international discussions, possibly incorporating alternative review methods like narrative literature reviews.

CRediT authorship contribution statement

Longmiao Gao: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yan Liu:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization. **Fiona X. Yang:** Writing – review & editing, Supervision. **Huimin Liu:** Writing – review & editing,

Supervision. **Ying Wang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yan Liu reports financial support was provided by National Natural Science Foundation of China. Huimin Liu reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant No. 72201125 and 72471110).

Data availability

Data will be made available on request.

References

- Azzali, S. (2019). Mega sporting events and their impact on the built environment: Lessons learned from the past. A| Z ITU Journal of the Faculty of Architecture, 16, 25–37.
- Bernardino, S., Freitas Santos, J., & Cadima Ribeiro, J. (2018). The legacy of European Capitals of Culture to the “smartness” of cities: The case of Guimarães 2012. *Journal of Convention & Event Tourism*, 19, 138–166.
- Bovy, P. (2006). Solving outstanding mega-event transport challenges: The Olympic experience. *Public Transport International*, 6, 32–34.
- Boyle, P. (2012). Risk, resiliency, and urban governance: The case of the 2010 Winter Olympic Games. *Canadian Review of Sociology/Revue canadienne de sociologie*, 49, 350–369.
- Chen, G., Peachey, J. W., Stodolska, M., Hooimeijer, P., & Lin, Y. (2021). Governing adaptation strategies of winter tourism destinations in the context of the 2022 Olympic Winter Games. *Tourism Review International*, 25, 1–17.
- Chopra, S. S., Dillon, T., Bilec, M. M., & Khanna, V. (2016). A network-based framework for assessing infrastructure resilience: A case study of the London metro system. *Journal of the Royal Society Interface*, 13, Article 20160113.
- Clark, J., Kearns, A., & Cleland, C. (2016). Spatial scale, time and process in mega-events: The complexity of host community perspectives on neighborhood change. *Cities*, 53, 87–97.
- Coaffee, J. (2021). Futureproofing against shock: Institutional responses to terrorist risk in London, 1990–2020. *Applied Geography*, 133, Article 102493.
- Coaffee, J., Fussey, P., & Moore, C. (2011). Laminated security for London 2012: Enhancing security infrastructures to defend mega sporting events. *Urban Studies*, 48, 3311–3327.
- Cronin, P., Ryan, F., & Coughlan, M. (2008). Undertaking a literature review: A step-by-step approach. *British Journal of Nursing*, 17, 38–43.
- Darnell, S. C. (2012). Olympism in action, Olympic hosting and the politics of ‘sport for development and peace’: Investigating the development discourses of Rio 2016. *Sport in Society*, 15, 869–887.
- De Angelis, M. C., Borrelli, I. P., Viola, I., Misso, R., & Cesaretti, G. P. (2016). Expo 2015 and the fight against climate change: Possible compatibility and strategies. *Calitatea*, 17, 555.
- Deakin, M., Diamantini, D., & Borrelli, N. (2019). The governance of a smart city food system: The 2015 Milan World Expo. *City, Culture and Society*, 16, 5–11.
- Deng, H., Mancini, M., Zhang, L., & Bonino, M. (2020). Beijing 2022 between urban renovation and Olympic sporting legacy: The case of Shougang. *Mov Sport Sci/Sociomot*, 53–65.
- Deng, Y., Poon, S. W., & Chan, E. H. W. (2016). Planning mega-event built legacies - A case of Expo 2010. *Habitat International*, 53, 163–177.
- Draca, M., Machin, S., & Witt, R. (2011). Panic on the streets of London: Police, crime, and the July 2005 terror attacks. *American Economic Review*, 101, 2157–2181.
- Duignan, M. B., Everett, S., & McCabe, S. (2022). Events as catalysts for communal resistance to overtourism. *Annals of Tourism Research*, 96, Article 103438.
- Edizel, Ö. (2013). Mega-events as a place marketing strategy in entrepreneurial cities: İzmir’s EXPO 2015 candidacy as a roadmap for hosting EXPO 2020. *Town Planning Review*, 633–657.
- Feng, M., Ren, J., He, J., Chan, F. K. S., & Wu, C. (2022). Potency of the pandemic on air quality: An urban resilience perspective. *Science of the Total Environment*, 805, Article 150248.
- Ferranti, E., Andres, L., Denoon-Stevens, S. P., Melgaço, L., Oberling, D., & Quinn, A. (2020). Operational challenges and mega sporting events legacy: The case of BRT systems in the Global South. *Sustainability*, 12, 1609.

- Ferranti, E. J. S., Oberling, D. F., & Quinn, A. D. (2022). Transport resilience to weather and climate: An interdisciplinary view from Rio de Janeiro. *Proceedings of the Institution of Civil Engineers-Urban Design and Planning*, 175, 103–121.
- Flax, L., Korthals Altes, R., Kupers, R., & Mons, B. (2020). Greening schoolyards - An urban resilience perspective. *Cities*, 106, Article 102890.
- Flyvbjerg, B., Stewart, A., & Budzier, A. *The Oxford Olympics Study 2016: Cost and cost overrun at the games*. (2016). arXiv preprint. arXiv:1607.04484.
- Fröhlich, K., & Hassink, R. (2018). Regional resilience: A stretched concept? *European Planning Studies*, 26, 1763–1778.
- Gaeta, L., & Vita, S. D. (2021). Planning disaster, successful event, and uncertain future: The twin cases of the World Expo 2015 and Innovation District in Milan. In *Les Cahiers de la recherche architecturale urbaine et paysagère*.
- Gezici, F., & Er, S. (2014). What has been left after hosting the Formula 1 Grand Prix in Istanbul? *Cities*, 41, 44–53.
- Girgin, F. A., & Tasci, O. E. (2019). Mega-event organization considering safety, security and resilience. *TEMA Journal of Land Use, Mobility and Environment*, 12, 249–264.
- Gollagher, P., & Fastenrath, S. (2023). Transformative climate resilience and sport mega-events – The case of the Australian Open. *Environmental Innovation and Societal Transitions*, 48, Article 100762.
- Goodwin, D., Raffin, M., Jeffrey, P., & Smith, H. M. (2019). Collaboration on risk management: The governance of a non-potable water reuse scheme in London. *Journal of Hydrology*, 573, 1087–1095.
- Gospodini, A. (2009). Post-industrial trajectories of Mediterranean European cities: The case of post-Olympics Athens. *Urban Studies*, 46, 1157–1186.
- Grabher, G., & Thiel, J. (2015). Projects, people, professions: Trajectories of learning through a mega-event (the London 2012 case). *Geoforum*, 65, 328–337.
- Gu, C., & Cook, I. G. (2012). Beijing: Socialist Chinese capital and new world city. In *Planning Asian cities*. Routledge.
- Gundes, S. (2022). Trends in global infrastructure investment and financial consequences. *European Journal of Sustainable Development*, 11, 66.
- Hassink, R. (2010). Regional resilience: A promising concept to explain differences in regional economic adaptability? *Cambridge Journal of Regions, Economy and Society*, 3, 45–58.
- Horne, J., & Manzenreiter, W. (2006). An introduction to the sociology of sports mega-events. *The Sociological Review*, 54, 1–24.
- Kassens-Noor, E., & Lauermann, J. (2017). How to bid better for the Olympics: A participatory mega-event planning strategy for local legacies. *Journal of the American Planning Association*, 83, 335–345.
- Kobierecki, M. M., & Pierzgalski, M. (2022). Sports mega-events and economic growth: A synthetic control approach. *Journal of Sports Economics*, 23, 567–597.
- Kwiatkowski, G., Ossowska, L., Strzelecka, M., Dragin-Jensen, C., Hannevik Lien, V., Janiszewska, D., & Klosowski, D. (2023). Building a resilient event sector in times of uncertainty. *Event Management*, 27, 481–498.
- Lauermann, J. (2016). Municipal statecraft: Revisiting the geographies of the entrepreneurial city. *Progress in Human Geography*, 42, 205–224.
- Lee, H., & Hlee, S. (2021). The intra- and inter-regional economic effects of smart tourism city Seoul: Analysis using an input-output model. *Sustainability*, 13, 4031.
- Lee, J. W. (2019). A winter sport mega-event and its aftermath: A critical review of post-Olympic PyeongChang. *Local Economy*, 34, 745–752.
- Lee Ludvigsen, J. A. (2021). When 'the show' cannot go on: An investigation into sports mega-events and responses during the pandemic crisis. *International Review for the Sociology of Sport*, 57, 497–514.
- Lenskyj, H. J. (2012). The Winter Olympics: Geography is destiny? In H. J. Lenskyj, & S. Wagg (Eds.), *The Palgrave handbook of Olympic studies*. London: Palgrave Macmillan UK.
- Li, L. (2021). Contesting sustainability of mega-events in Chinese metropolises: A narrative and practise review. *Frontiers in Sustainable Cities*, 3.
- Lin, G. C. S., He, C., Li, X., & Wu, Y. (2018). Empowering regional economy with a spectacular space: Mega-events, over-drafted capital and momentary growth in China's metropolises. *Area Development and Policy*, 3, 24–41.
- Majoor, S. (2015). Resilient practices: A paradox-oriented approach for large-scale development projects. *Town Planning Review*, 86, 257–277.
- Meerow, S., & Newell, J. P. (2019). Urban resilience for whom, what, when, where, and why? *Urban Geography*, 40, 309–329.
- Meerow, S., Newell, J. P., & Stults, M. (2016). Defining urban resilience: A review. *Landscape and Urban Planning*, 147, 38–49.
- Meza Talavera, A., Al-Ghamdi, S. G., & KOÇ, M. (2019). Sustainability in mega-events: Beyond Qatar 2022. *Sustainability*, 11, 6407.
- Michelam, L. D., Cortese, T. T. P., Yigitcanlar, T., Fachinelli, A. C., Vils, L., & Levy, W. (2021). Leveraging smart and sustainable development via international events: Insights from Bento Gonçalves Knowledge Cities World Summit. *Sustainability*, 13, 9937.
- Moher, D., Shamseer, L., Clarke, M., Ghersi, D., Liberati, A., Petticrew, M., ... Group, P.-P. (2015). Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews*, 4, 1–9.
- Müller, M. (2015). The mega-event syndrome: Why so much goes wrong in mega-event planning and what to do about it. *Journal of the American Planning Association*, 81, 6–17.
- Müller, M., Wolfe, S. D., Gaffney, C., Gogishvili, D., Hug, M., & Leick, A. (2021). An evaluation of the sustainability of the Olympic Games. *Nature Sustainability*, 4, 340–348.
- Nakamura, H., & Suzuki, N. (2017). Reinterpreting Olympic legacies: The emergent process of long-term post-event strategic planning of Hakuba after the 1998 Nagano Winter Games. *International Journal of Sport Policy and Politics*, 9, 311–330.
- Nguyen, D. N., Esteban, M., & Onuki, M. (2019). Resiliency in tourism transportation: Case studies of Japanese railway companies preparing for the 2020 Tokyo Olympics. *International Journal of Disaster Risk Reduction*, 38.
- Orr, M., Inoue, Y., Seymour, R., & Dingle, G. (2022). Impacts of climate change on organized sport: A scoping review. In *Wiley interdisciplinary reviews-Climate change*.
- Pereira, R. H. M. (2018). Transport legacy of mega-events and the redistribution of accessibility to urban destinations. *CITIES*, 81, 45–60.
- Ponzini, D. (2022). Introduction to the special issue – Cultural mega-events and heritage: Challenges for European cities. *European Planning Studies*, 30, 427–436.
- Quinlan, A. E., Berbés-Blázquez, M., Haider, L. J., & Peterson, G. D. (2016). Measuring and assessing resilience: Broadening understanding through multiple disciplinary perspectives. *Journal of Applied Ecology*, 53, 677–687.
- Raco, M., & Tunney, E. (2010). Visibilities and invisibilities in urban development: Small business communities and the London Olympics 2012. *Urban Studies*, 47, 2069–2091.
- Ramos, S. P., & Mundet, L. (2021). Tourism-phobia in Barcelona: Dismantling discursive strategies and power games in the construction of a sustainable tourist city. *Journal of Tourism and Cultural Change*, 19, 113–131.
- Rekow, L. (2015). Fighting insecurity: Experiments in urban agriculture in the favelas of Rio de Janeiro. *Field Actions Science Reports. The Journal of Field Actions*, 8.
- Richards, G., & Wilson, J. (2004). The impact of cultural events on city image: Rotterdam, cultural capital of Europe 2001. *Urban Studies*, 41, 1931–1951.
- Robinson, J. A., & Torvik, R. (2005). White elephants. *Journal of Public Economics*, 89, 197–210.
- Rocha, C. M. (2023). Rio 2016 Olympic legacy for residents of favelas: Revisiting the case of Vila Autódromo five years later. *Social Sciences-Basel*, 12.
- Rocha, C. M., & Xiao, Z. (2022). Sport mega-events and displacement of host community residents: A systematic review. *Frontiers in Sports and Active Living*, 3, 2021.
- Rogerson, R. J. (2016). Re-defining temporal notions of event legacy: Lessons from Glasgow's Commonwealth Games. *Annals of Leisure Research*, 19, 497–518.
- Ross, W. J., & Orr, M. (2022). Predicting climate impacts to the Olympic Games and FIFA Men's World Cups from 2022 to 2032. *Sport in Society*, 25, 867–888.
- Salvati, L., & Serra, P. (2016). Estimating rapidity of change in complex urban systems: A multidimensional, local-scale approach. *Geographical Analysis*, 48, 132–156.
- Scott, D., Steiger, R., Rutty, M., & Fang, Y. (2019). The changing geography of the Winter Olympic and Paralympic Games in a warmer world. *Current Issues in Tourism*, 22, 1301–1311.
- Scott, D., Steiger, R., Rutty, M., & Johnson, P. (2015). The future of the Olympic Winter Games in an era of climate change. *Current Issues in Tourism*, 18, 913–930.
- Serdar, M. Z., & Al-Ghamdi, S. G. (2021). Resiliency assessment of road networks during mega sport events: The case of FIFA World Cup Qatar 2022. *Sustainability*, 13, Article 12367.
- Serdar, M. Z., & Al-Ghamdi, S. G. (2023). Resilience-oriented recovery of flooded road networks during mega-sport events: A novel framework. *Frontiers in Built Environment*, 9.
- Shin, H. B. (2014). Urban spatial restructuring, event-led development and scalar politics. *Urban Studies*, 51, 2961–2978.
- Shin, N., & Welty Peachey, J. (2022). Understanding the global-local nexus in the context of the Olympic Games: Implications for managing community development through sport mega-events. *Journal of Sport Management*, 36, 82–95.
- Shipway, R., & Miles, L. (2020). Bouncing back and jumping forward: Scoping the resilience landscape of international sports events and implications for events and festivals. *Event Management*, 24, 185–196.
- Silver, C. (2014). Spatial planning for sustainable development: An action planning approach for Jakarta. *Journal of Regional and City Planning*, 25, 115–125.
- Talbot, A. (2023). Human rights at the Olympic Games: Policy, protest, progress? *Event Management*, 27, 915–930.
- Thompson, C., Lewis, D., Greenhalgh, T., Taylor, S., & Cummins, S. (2013). A health and social legacy for East London: Narratives of 'problem' and 'solution' around London 2012. *Sociological Research Online*, 18, 144–149.
- Thomson, A., Cuskelly, G., Toohey, K., Kennelly, M., Burton, P., & Fredline, L. (2019). Sport event legacy: A systematic quantitative review of literature. *Sport Management Review*, 22, 295–321.
- Toohey, K., & Taylor, T. (2008). Mega events, fear, and risk: Terrorism at the Olympic Games. *Journal of Sport Management*, 22, 451–469.
- Walker, B. (2020). Resilience: What it is and is not. *Ecology and Society*, 25, 11.
- Wang, M., Bao, H. X. H., & Lin, P. T. (2015). Behavioural insights into housing relocation decisions: The effects of the Beijing Olympics. *Habitat International*, 47, 20–28.
- Wang, Y., & Jin, X. (2019). Event-based destination marketing: The role of mega-events. *Event Management*, 23, 109–118.
- Weichselgartner, J., & Kelman, I. (2015). Geographies of resilience: Challenges and opportunities of a descriptive concept. *Progress in Human Geography*, 39, 249–267.
- Yamamura, E. (2021). Do you want sustainable Olympics? Environment, disaster, gender, and the 2020 Tokyo Olympics. *Sustainability*, 13, Article 12879.
- Zheng, M., Cui, N. N., Zhang, Y. B., Zhang, F. F., & SHI, V. (2023). Inventory policies and supply chain coordination under logistics route disruption risks. *Sustainability*, 15.